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Substrate processing apparatus, reactor mover for moving a reactor of the apparatus and method of maintaining the reactor of the apparatus

Abstract

The disclosure relates to a substrate processing apparatus for processing a plurality of substrates. The apparatus comprising a reactor mounted in the apparatus and configured for processing substrates and a reactor mover for moving the reactor for maintenance. The reactor mover is constructed and arranged with a lift to move the reactor to a lower height to allow for access to the reactor by a maintenance worker.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/035,208 filed Jun. 5, 2020 titled “SUBSTRATE PROCESSING APPARATUS, REACTOR MOVER FOR MOVING A REACTOR OF THE APPARATUS AND METHOD OF MAINTAINING THE REACTOR OF THE APPARATUS,” the disclosure of which is hereby incorporated by reference in its entirety.

FIELD

(1) The present disclosure generally relates to a substrate processing apparatus for processing a plurality of substrates. More particularly, the disclosure relates to a substrate processing apparatus for processing a plurality of substrates, comprising: a reactor mounted in the apparatus and configured for processing substrates and a reactor mover for moving the reactor for maintenance.

BACKGROUND

(2) For feeding cassettes which accommodate a plurality of substrates to the substrate processing apparatus a cassette handler may transfer substrate cassettes between a cassette in-out port, a substrate transfer position and/or a storage device. The storage device may comprise a cassette storage carousel for storing a plurality of cassettes. The substrate handling robot may be constructed and arranged to transfer substrates between a cassette at the substrate transfer position and a substrate rack in a lower region of the apparatus.

(3) The lower region may, optionally, be provided with a rack conveyor which may include a carousel or rotatable arm on which the rack may be positioned. By rotating the carousel or rotatable arm, the rack may be brought from an unloading and/or loading position to a treatment position under the first or second reactor.

(4) An elevator may be provided in the lower region to lift the rack with substrates into the reactor or to lower a rack with processed substrates from the reactor. During treatment in the reactor, the rack and the substrates accommodated therein may be raised in temperature. With the lift, the carousel or rotatable arm, a substrate rack may be removed from the reactor to cool down after treatment.

(5) Substrate processing apparatus may have a reactor that may require maintenance. For example, the reactor may comprise components like an insulation material, a heating element, a temperature sensor, a tube, a liner, an injector, an exhaust, a flange, a rack, and/or a pedestal that may get contaminated, broken or just simply needs to be altered during maintenance. Since these components may be very heavy and mounted in the apparatus at a substantial height within the apparatus, they may not be easily accessible for maintenance by a maintenance worker.

SUMMARY

(6) This summary is provided to introduce a selection of concepts in a simplified form. These concepts are described in further detail in the detailed description of example embodiments of the disclosure below. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

(7) According to an objective it may be desirable to provide an apparatus which may be easier to maintain by a maintenance worker.

(8) Accordingly, there may be provided a substrate processing apparatus for processing a plurality of substrates, comprising: a reactor mounted in the apparatus and configured for processing substrates; and, a reactor mover for moving the reactor for maintenance. The reactor mover may be constructed and arranged with a lift to move the reactor to a lower height to allow for access to the reactor by a maintenance worker.

(9) According to a further embodiment, there may be provided a reactor mover for moving a reactor of a substrate processing apparatus for maintenance, wherein the reactor mover comprises a lift and a displacer constructed and arranged to: move the reactor to a maintenance area outside the apparatus with the displacer; and, lower the reactor in the maintenance area outside the housing with the lift.

(10) According to a further embodiment, there may be provided a method of maintaining a reactor a substrate processing apparatus by: moving the reactor to a maintenance area outside the apparatus with a displacer; lowering the reactor in the maintenance area outside the housing with the lift; and, providing maintenance on the reactor.

(11) For purposes of summarizing the invention and the advantages achieved over the prior art,

certain objects and advantages of the invention have been described herein above. Of course, it is to be understood that not necessarily all such objects or advantages may be achieved in accordance with any particular embodiment of the invention. Thus, for example, those skilled in the art will recognize that the invention may be embodied or carried out in a manner that achieves or optimizes one advantage or group of advantages as taught or suggested herein without necessarily achieving other objects or advantages as may be taught or suggested herein.
(12) All of these embodiments are intended to be within the scope of the invention herein disclosed. These and other embodiments will become readily apparent to those skilled in the art from the following detailed description of certain embodiments having reference to the attached figures, the invention not being limited to any particular embodiment(s) disclosed.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) While the specification concludes with claims particularly pointing out and distinctly claiming what are regarded as embodiments of the invention, the advantages of embodiments of the disclosure may be more readily ascertained from the description of certain examples of the embodiments of the disclosure when read in conjunction with the accompanying drawings, in which:
(2) FIG. **1** shows a schematic top view of an example of the substrate processing apparatus according to an embodiment.
(3) FIG. **2**A shows a schematic top view of a portion of the substrate processing apparatus of FIG. **1** during maintenance to a reactor.
(4) FIG. **2**B shows a schematic top view of a portion of the substrate processing apparatus of FIG. **1** during replacement of a reactor.
(5) FIG. **2**C shows a schematic top view on reactor mover for moving the reactor for maintenance according to an embodiment.
(6) FIG. **2**D shows a perspective side view on reactor mover for moving the reactor for maintenance according to an embodiment.
(7) FIG. **2**E shows a perspective side view on reactor mover for moving the reactor for maintenance according to an embodiment.
(8) FIGS. **3**A and **3**B show a side view of a reactor move out of a substrate processing apparatus during maintenance according to an embodiment.
(9) FIGS. **4**A, B, and C show a side view of a reactor move out of a substrate processing apparatus during maintenance according to an embodiment.
(10) FIGS. **5**A and B show a side view of a reactor move out of a substrate processing apparatus during maintenance according to an embodiment.
(11) FIGS. **6**A, B and C show a side view of a reactor move out of a substrate processing apparatus during maintenance according to an embodiment.
DETAILED DESCRIPTION OF THE FIGURES
(12) Although certain embodiments and examples are disclosed below, it will be understood by those in the art that the invention extends beyond the specifically disclosed embodiments and/or uses of the invention and obvious modifications and equivalents thereof. Thus, it is intended that the scope of the invention disclosed should not be limited by the particular disclosed embodiments described below. The illustrations presented herein are not meant to be actual views of any particular material, structure, or device, but are merely idealized representations that are used to describe embodiments of the disclosure.
(13) As used herein, the term “substrate” or “wafer” may refer to any underlying material or materials that may be used, or upon which, a device, a circuit, or a film may be formed. The term

"semiconductor device structure" may refer to any portion of a processed, or partially processed, semiconductor structure that is, includes, or defines at least a portion of an active or passive component of a semiconductor device to be formed on or in a semiconductor substrate. For example, semiconductor device structures may include, active and passive components of integrated circuits, such as, for example, transistors, memory elements, transducers, capacitors, resistors, conductive lines, conductive vias, and conductive contact pads.
(14) FIG. **1** shows a schematic top view of an example of the substrate processing apparatus **1** according to an embodiment. The substrate processing apparatus **1** comprises a housing **2** with a front wall **4** and a back wall **6**.
(15) The substrate processing apparatus **1** may comprise a storage device such as a cassette storage carousel **3** for storing a plurality of wafer cassettes C which wafer cassettes each accommodate a plurality of substrates. The cassette storage carousel **3** may comprise a number of platform stages for supporting cassettes. The platform stages may be connected to a central support which is mounted rotatable around a vertical axis **5**. Each platform stage may be configured for accommodating a number of cassettes C. A drive assembly may be operatively connected to the central support for rotating the central support with the platform stages around the vertical axis **5**.
(16) The substrate processing apparatus **1** may have a cassette handler **7** having a cassette handler arm **9** configured to transfer cassettes C between the cassette storage carousel **3**, a cassette in-out port **11** adjacent the front wall **4** of the housing **2** of the substrate processing apparatus **1**, and/or a substrate transfer position **15**. The cassette handler **7** may comprise an elevating mechanism to reach to the cassettes at different height. Each platform stage for storing cassettes may have a cut-out therein the cut-out sized and shaped to allow the cassette handler arm **9** to pass vertically there through and to allow the platform stage to support a cassette C thereon.
(17) An internal wall **19** separating the cassette handler **7** and the substrate handling robot **35** may be provided. The wall may have a closable substrate access opening **33** adjacent the substrate transfer position **15** which may be constructed and arranged to also open the cassette. More information with respect to the closeable substrate access opening **33** for opening cassettes may be gleaned from U.S. Pat. No. 6,481,945, which is incorporated herein by reference. The substrate transfer position **15** may be provided with a cassette turntable to turn the cassette C and/or to press it against the closeable substrate access opening **33**.
(18) The substrate processing apparatus may comprise a substrate handling robot **35** provided with a substrate handling arm **36** to transfer substrates from a cassette C positioned on the substrate transfer position **15** through the closeable substrate access opening **33** to a substrate rack and vice versa. The substrate processing apparatus may comprise a substrate handling chamber **37** in which the substrate handling robot **35** is accommodated.
(19) The housing **2** may have a first and second side wall **47** extending over the full length of the substrate processing apparatus **1**. The distance between the sidewalls **47** of the substrate processing apparatus which may define the width of the apparatus may be between 190 and 250 cm, preferably between 210 and 230 cm and most preferably around 220 cm. Maintenance of the substrate processing apparatus **1** may be performed from the backside **6** or front side **4** of the substrate processing apparatus so that there may be no need for doors in the sidewalls **47**.
(20) With the construction of the sidewalls **47** without doors multiple substrate processing apparatus **1** may be positioned side by side in a semiconductor fabrication plant. The sidewall of adjacent substrate processing apparatus may thereby be positioned very close together, or even against each other. Advantageously, the multiple substrate processing apparatus may form a wall with the front side **4** of the substrate processing apparatus **1** interfacing with a cassette transport device in a very clean environment of a so called "clean room" having very strict requirements for particles. The backside **6** of the substrate processing apparatus **1** may interface with a maintenance alley which may have less strict requirements for particles than the front side **4**.
(21) The substrate processing apparatus **1** may be provided with a first and a second reactor **45** for

processing a plurality of substrates. Using two reactors may improve the productivity of the substrate processing apparatus **1**. The reactor **45** may comprise components like an insulation material, a heating element, a temperature sensor, a tube, a liner, an injector, an exhaust, a flange, a rack, and/or a pedestal.

(22) The substrate processing apparatus in top view may be configured in a substantial U-shape. The first and second reactors **45** may be constructed and arranged in the legs. The first and second reactors **45** may be provided in the lower part of the legs of the U-shape.

(23) A maintenance area **43** may be constructed and arranged between the legs of the U-shape. The distance between the two legs of the U-shape may be between 60 and 120 cm, more preferably between 80 and 100 cm, and most preferably around 90 cm to allow enough space for a maintenance worker. A back door **42** may be provided between the legs of the U-shape to close off the maintenance area **43** when not in use. The substantial U-shape may include a V-shape as well.

(24) FIG. **2**A shows a schematic top view of a back portion of the substrate processing apparatus of FIG. **1** during maintenance with the back doors **42** opened. The apparatus may be constructed and arranged to allow maintenance of the reactors **45** from the maintenance area **43** to both the first and second reactor by a maintenance worker M. For example, for maintenance on heating elements, heating sensors, and/or processing gas interfaces **65** (outlets and/or inlets) by the maintenance worker M. The two reactors **45** may be constructed and arranged adjacent to the same maintenance area **43** to allow maintenance of both reactors **45** from the same maintenance area **43**. A reactor door **41** may be opened to allow maintenance of the reactor **45**.

(25) A maintenance worker M therefore may not enter the housing **2** of the substrate processing apparatus **1** for maintenance work on the reactor **45**. Risk of injury of the maintenance worker M and/or contamination of the lower region and/or the reactor **45** may therefore be minimized. Maintenance from outside the housing may also improve the speed and/or accuracy of the maintenance. By using the same maintenance area for both reactors **45** the total footprint of the apparatus may be reduced.

(26) The reactors **45** may be provided in a treatment module **49** which also may include the lower region of the substrate processing apparatus **1** provided with an elevator configured to transfer a rack or boat with substrates between the lower region and the reactor **45**. The elevator may be constructed and arranged in the legs of the U-shape as well. A gate valve **63** may be provided between the substrate handler **35** and the lower region of the treatment modules **49** creating a mini environment in the lower region.

(27) A rack conveyor **55** for conveying the rack may be provided in the lower region of the substrate processing apparatus. The rack conveyor **55** may comprise two accommodations **57** for racks. Alternatively, the rack conveyor may comprise three or four accommodations for racks. The rack conveyor **55** may be constructed and arranged in the legs and bottom part of the U-shape. The rack conveyor **55** may be constructed and arranged to horizontally transfer racks between a plurality of rack positions within the lower region.

(28) Accommodation **57** may be a cooling down accommodation for cooling down a rack with just processed hot substrates. The cooling down accommodation may be provided with a gas inlet and outlet. The gas inlet and outlet may be fluidly connected to a pump to create a gas flow through the mini environment in the lower region.

(29) The module **49** may be provided with an elevator configured to transfer a rack with substrates between a lower region of the substrate processing apparatus and the reactor **45**. The elevator may comprise a rack support arm having a bearing surface configured to support a substrate rack. The rack conveyor **55** may have a support platform having a cut-out therein. The cut-outs may be sized and shaped to allow the bearing surface of the rack support arm to translate vertically there through and to allow the platform to support a substrate rack. The rack conveyers **55** may be rotatable to transport the rack horizontally from a position close to the substrate handling robot **35** to a position underneath the reactor **45**.

(30) The treatment module **49** may be provided with a reactor door **41** openable in a direction of the maintenance area **43**. Both treatment modules **49** may be provided with a separate reactor door **41** to allow maintenance of each reactor **55** from the same maintenance area **43** by the maintenance worker M. The reactor door **41** of the treatment module **49** may also allow maintenance of the lower region of the substrate processing apparatus from the same maintenance area **43** for maintenance on the rack conveyers **55**, gas inlet, gas outlet, gate valves **63** elevator and/or rack.
(31) Alternatively, two reactor doors may be provided per treatment module; a first reactor door for giving access to the reactor **45** and a second reactor door for giving access to the lower region of the substrate processing apparatus. For example, for maintenance on the rack conveyers **55**, gas inlet, gas outlet, gate valves **63** elevator and/or rack access to the lower region of the substrate processing apparatus may be required.
(32) A maintenance worker M therefore may not need to enter the housing **2** of the substrate processing apparatus **1** for maintenance work inside the treatment module **49**. Risk of injury of the maintenance worker M and/or contamination of the inside of the treatment module may thereby be minimized.
(33) Gate valves **63** may be provided between the substrate handling chamber **37** and the treatment modules **49**. The gate valves **63** may be closed during maintenance to the substrate handling robot **35** so that the treatment modules may continue working while maintaining the substrate handling robot **35**.
(34) Maintenance on one of the reactors **45** may not interfere with the other one of the reactors **45** since both are constructed as separate units. The gate valves **63** may be closed during maintenance of one reactor so that the other reactor and the substrate handling robot **35** may continue working.
(35) The reactors **45** may have processing gas interfaces **65** (outlets and/or inlets) which may be connected to a gas exhaust pipe or a process gas delivery system. The processing gas interfaces **65** may be provided adjacent the maintenance area **43** so as to make them easily accessible during maintenance.
(36) The process gas delivery system may be provided (partly) in a gas cabinet **67** constructed and arranged to provide process gas to the first and/or second reactor **45**. The gas cabinet may be provided near a top of the legs of the U-shape. Providing the gas cabinet near the top of the U-shape provides easy access for maintenance. Further this place provides flexibility if the gas cabinet needs to be adjusted or enlarged since there is space at the back of the apparatus where no critical components as substrate handlers or cassette handlers are present.
(37) The gas exhaust pipe may be constructed and arranged to remove process gas from at least one of the first and the second reactors and may also be provided near a top of the legs of the U-shape. Providing the gas exhaust pipe in the legs of the U-shape provides easy access for maintenance. The gas exhaust pipe may be provided with pumps and scrubbers which may require regular maintenance. Further this place provides flexibility if the gas exhaust pipe needs to be adjusted or extended since there is space at the back of the apparatus where no critical components such as substrate handling robots and/or cassette handlers are present.
(38) FIG. **2**B shows a schematic top view of a back portion of the substrate processing apparatus of FIG. **1** during maintenance with the back doors **42** opened. Maintenance to the reactors may include the replacement of large components such as reactors, reactor tubes, substrate racks, heating elements, doors, pumps and/or valves.
(39) A maintenance worker M standing on a maintenance area **43** outside the housing **2** may remove the reactor **45** for example from the apparatus. Such large components including the tooling to transport them safely may require a large maintenance area **43**. Since the same maintenance area **43** may be used for both reactors **45** the footprint of the maintenance area **43** may be reduced. The opening at the back side **6** of the apparatus easily allows the transportation of even big components, such as the reactor **45** out of the apparatus.
(40) FIG. **2**C shows a schematic top view on reactor mover **69** for moving the reactor **45** (in FIGS.

1 and **2**A, **2**B) for maintenance. The reactor mover **69** may be mounted at a first end **71** to a frame **100** of substrate processing apparatus **1** (in FIG. **1**) and at the other via a suspension frame, e.g., C-shape frame **92** to a reactor **45**. The C-shape frame **92** may be constructed and arranged to at least partially embrace the reactor **45**.

(41) The reactor mover **69** may be constructed and arranged with a displacer **73** to move the reactor **45** in a substantial horizontal direction to a maintenance area **43** (in FIG. **2**A). The maintenance area **43** may be outside a housing of the apparatus and the displacer **73** may move the reactor **45** there. The displacer **73** may be provided with a swing mechanism **94**.

(42) The swing mechanism **94** may be constructed and arranged to swing the reactor around a substantially vertical axis. The swing mechanism **94** may comprise a first hinge **76** and a swinging arm **77** swingable around the first substantially vertical axis. The swing mechanism **94** may comprise a second hinge **78** and the reactor may be supported by the swinging arm **77** via the second hinge **78** swingable around a second substantially vertical axis. The swing mechanism **94** may comprise a third hinge **79** and the reactor **45** may be supported by the swinging arm **77** via the third hinge swingable around a third substantially vertical axis.

(43) The construction with three hinges **76**, **78**, **79** and their respective vertical axes makes that the reactor can be moved relatively freely in the horizontal plane so that the reactor can be positioned at the right position on the flange **25**. This movement may be accomplished by hand or motor **75** driven.

(44) FIG. **2**D shows a perspective side view on reactor mover **69** for moving the reactor **45** (in FIGS. **1** and **2**A, **2**B) for maintenance. The reactor mover **69** may be constructed and arranged with a lift **80** to move the reactor up and down. The lift **80** may be used to bring the reactor to a lower height to allow for access to the reactor by a maintenance worker.

(45) The lift **80** may be provided as a linear guide mechanism comprising a linear guide **81** in the second hinge **78**. The linear guide mechanism may guide the reactor in a substantial vertical direction over a substantially vertical linear guide **81**. The C-shape frame **92** may form part of a carriage **91** moveable along the linear guide **81** via the third hinge **79**.

(46) The carriage **91** and therefore the C-shape frame **92** may be driven up and down by a threaded pin **82** in a rotatable nut **83**. A motor or a crank handle **84** may rotate the rotatable nut **83** via a transmission and move the C-shape frame up and down over the linear guide **81**. In this way the lift **80** may be formed in the swing mechanism **94** to move the reactor in a substantial vertical direction in a compact design.

(47) A part of the swinging arm **77** connects the first hinge **76** and the second hinge **78**. The lift **80** may be constructed and arranged to move the second hinge **78** relative to the first hinge **76**.

(48) The reactor mover **69** may move the reactor up within the housing of the apparatus **1** with the lift **80**, may move the reactor to a maintenance area **43** outside the housing of the apparatus **1** with the displacer **73**, and, may lower the reactor **45** in the maintenance area **43** outside the housing with the lift **80**.

(49) Alternatively, a linear guide mechanism forming part of a displacer may move the reactor in a substantial horizontal direction to a maintenance area over a substantially horizontal linear guide.

(50) The substrate processing apparatus **1** may be provided with two reactors **45** and a single reactor mover **69** may be constructed and arranged to move each reactor **45** to a maintenance area **43**. The maintenance area **43** may be the same maintenance area **43** for both reactors **45**. The maintenance area may be provided in between the first and second reactor. The apparatus may be provided with two reactor movers **69** each constructed and arranged to move one of the first and second reactor **45** to the maintenance area **43**.

(51) FIG. **2**E shows a perspective side view on reactor mover **69** for moving the reactor **45** (in FIGS. **1** and **2**A, **2**B and **2**C) for maintenance. The reactor bottom support **87** may be part of the reactor **45** (not shown). The reactor bottom support **87** may support the rest of the reactor **45** sitting on top of it. The reactor bottom support **87** may be provided with an air circulation duct **86**.

(52) The reactor mover may be provided with a suspension frame and the reactor **45** may be suspended from the suspension frame with elongated suspensions **85**. The reactor bottom support **87** may be suspended from the reactor mover **69** via elongated suspensions **85** during lifting and moving of the reactor **45**. More in particular the bottom support **87** may be suspended from the reactor mover **69** via the suspension frame, e.g., C-shape frame **92** and three elongated suspensions **85**, which may be a cable or rod. The elongated suspensions **85** may be provided with cable tensioners **89**, e.g., turnbuckles to adjust the position of the reactor **45** in the reactor mover during lifting.
(53) The C-shape frame **92** and the three elongated suspensions **85** create room for a substantial part of the reactor **45** to hang there in between. The center of weight of the reactor **45** may be between the suspensions **85**. To accomplish that, the C-shape frame may therefore embrace the reactor **45** more than 180 degrees around, for example 200 degrees around. The C-shape frame **92** may still having an opening at the open side of the C-shape frame **92** that allows (portions of) the reactor **45** to move in and be mounted on the bottom support **87**. The elongated suspensions may be aligned with a vertical direction. The suspensions **85** may only experience tensile stress.
(54) The three suspensions **85** allow for the reactor **45** to settle itself on a flange **25** when the lift **80** of FIG. **2**D lowers the reactor on the flange. The tension on the suspensions **85** may then be released by lowering the C-shape frame **92** with the lift a little further. The suspensions **85** may be connected with flexibility to the C-shape frame **92** and the reactor bottom support **87**.
(55) FIGS. **3**A and **3**B show a side view of a reactor **45** move out of the substrate processing apparatus during maintenance. In FIG. **3**A, in step **93**, the reactor **45** may be moved away from a position above mini-environment **70** of the lower region into maintenance area **43** so that it can be maintained. This may be done by, for instance, with a reactor mover provided with a lift and displacer. As shown, a utilities cable **29**, e.g., for electricity or cooling water may connect the reactor **45** to the rest of the apparatus and may allow for this movement by having an additional length.
(56) The lift may be used to move the reactor **45** to a lower height to allow for access to the reactor by a maintenance worker M as shown in FIG. **3**B. The lift may also be used to move the reactor higher before moving it out of the apparatus so as to disengage it from its support. The displacer may comprise a linear guide mechanism or a swing mechanism **94** to move the reactor **45** in a horizontal direction to the maintenance area **43**. The maintenance area **43** may be outside the housing of the processing apparatus. A door in the housing may be opened to allow for the reactor **45** to be moved outside the housing.
(57) As shown in FIG. **3**B, a maintenance worker M may perform maintenance actions such as removing an insulation material, a heating element **23**, a temperature sensor, a tube **24**, a flange **26** from the reactor **45**. To help the maintenance worker maintain the apparatus, the lift may be used to lower **98** the reactor **45** to a height convenient for maintenance.
(58) The maintenance worker M may remove in **95** a rack **96** from mini-environment **70**. It will be clear that for the removal of a rack **96** it may not be necessary that the reactor **45** is in a moved away position but may still be in its processing position above mini-environment **70**. For the removal of tube **24** a further tool may be used. It may be clear that not only the tube **24** may be removed but also flange **25** and optionally also heating element **23**.
(59) FIGS. A to C show a side view of a reactor move out of a substrate processing apparatus during maintenance according to an embodiment. FIG. **4** may show in further detail a maintenance operation that may be performed in connection with the system as described in FIGS. **1** to **2**.
(60) In FIG. **4** A, the reactor **45** is in process position and is shown for simplicity without heating element **23**. In this embodiment, the reactor may be **45** provided with a tube **24***a* and a liner **24***b*. The tube **24***a* and the liner **24***b* may be supported on one or more flanges **25**.
(61) This configuration may be used for low pressure applications wherein the process gases flow in upward direction within inner tube **24***a* and in downward direction, towards an exhaust in the

space between inner process tube **24**_b_ and outer process tube **24**_a_. An exhaust **68** may be configured in communication with a vacuum pump for exhausting the process gases.

(62) A rack **26**, supported on pedestal **27**, door plate **28** and lower plate **72** may be positioned in mini environment **70**, enclosed by a housing **70**_a_. An elevator, not shown, may lift the lower plate **72**, including door plate **28**, pedestal **27** and rack **26** to an uppermost position wherein lower plate **72** seals against a top surface **74** of mini-environment **70** and simultaneously door plate **28** seals against flanges **25**, as shown in FIG. **4** B. Further, door plate **28** may be secured against flanges **25** with locking means, not shown.

(63) Then, as shown in FIG. **4** C, the reactor **45** may in its entirety be elevated **97**_a_, optionally after disconnection of gas, vacuum and other tubing and connections, as needed by a lift (not shown). Finally, the reactor **45** may be moved horizontally **97**_b_ by a displacer **73** away from a position above mini-environment **70**, together with one or more of the flanges **25**, tube **24**_a_ and liner **24**_b_, exhaust **68**, rack **26**, pedestal **27** and door plate **28** to allow the performance of maintenance actions. The lift and displacer may be part of the reactor mover. During the removal of the reactor, the mini-environment **70** may be sealed by lower plate **72**, while the tube **24** may be sealed by door plate **28**, so that contamination during the maintenance operation is avoided.

(64) Moving the reactor **45** with rack **26**, pedestal **27** and door plate **28** makes it possible to exchange the rack **26** and the pedestal **27** at a location outside the apparatus. The latter may be advantageously if there may be limited access to the mini environment **70** or if the mini environment preferably is not opened to keep it clean.

(65) FIGS. **5** A and B show a side view of a reactor move out of a substrate processing apparatus during maintenance according to an embodiment. In FIG. **5** A, the reactor **45** is in process position and is shown for simplicity without heating element **23**. In this embodiment, the reactor is **45** provided with a tube **24**_a_ and a liner **24**_b_. The tube **24**_a_ and the liner **24**_b_ may be supported on one or more flanges **25**_a_, **25**_b_.

(66) As shown in FIG. **5**B the reactor **45** may be lifted with a lift (not shown) together with the first flange **25**_a_ which may be mounted to tube **24**_a_ by a mounting mechanism (not shown). This mounting mechanism may be provided by the maintenance worker or a permanent mount to connect the first flange with the tube may be present. The second flange **25**_b_ may remain on the minienvironment **70**.

(67) Moving the reactor **45** with first flange **25**_a_ may make it possible to exchange the first flange **25**_a_ at a location outside the apparatus. Also it may become possible to provide maintenance to components supported on the first flange **25**_a_ such as for example the liner **24**_b_ and/or an injector (not shown). The second flange **25**_b_ may stay in the apparatus on the minienvironments so that the exhaust that is connected to the second flange **25**_b_ doesn't need to be disconnected for the removal of the reactor **45**.

(68) FIGS. **6**A, B, and C show a side view of a reactor move out of a substrate processing apparatus during maintenance according to an embodiment. In FIG. **6**A, the reactor **45** is in the process position and is again shown for simplicity without heating element **23**. In this embodiment, the reactor **45** provided with a tube **24**_a_ and a liner **24**_b_ may be also supported on one or more flanges **25**_a_, **25**_b_.

(69) A shown in FIG. **6**B the reactor **45** may be lifted by a lift together with both flanges **25**_a_ and **25**_b_. A clamp mechanism may be used to clamp one or more flanges to the reactor **45**. More specifically the second flange **25**_b_ may be clamped to the first flange **25**_a_ by clamp **30** functioning as the clamping mechanism. The clamp **30** may for this purpose be provided by the maintenance worker for a move out of the second flange **25**_b_. The clamp **30** may also be permanently installed in the apparatus so that it may be released by a maintenance worker if the first flange **25**_a_ is to be moved out separately of the second flange **25**_b_ as in FIG. **5**.

(70) Moving the reactor **45** with the flanges **25**_a_, **25**_b_ with a displacer **73**, as shown in FIG. **6** C, makes it possible to provide maintenance to the second flange **25**_b_ outside the apparatus. For

example, the second flange **25***b* and the exhaust connected thereto may be cleaned or even completely exchanged. The exhaust may need to be disconnected from the rest of the apparatus if the second flange **25***b* is moved out. The lift and displacer may be part of the reactor mover.
(71) Although illustrative embodiments of the present invention have been described above, in part with reference to the accompanying drawings, it is to be understood that the invention is not limited to these embodiments. Variations to the disclosed embodiments can be understood and effected by those skilled in the art in practicing the claimed invention, from a study of the drawings, the disclosure, and the appended claims.
(72) Reference throughout this specification to "one embodiment" or "an embodiment" means that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of the phrases "in one embodiment" or "in an embodiment" in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, it is noted that particular features, structures, or characteristics of one or more embodiments may be combined in any suitable manner to form new, not explicitly described embodiments.

Claims

1. A substrate processing apparatus for processing a plurality of substrates, comprising: a frame; a reactor mounted to the frame of the apparatus and configured for processing substrates; and, a reactor mover for moving the reactor for maintenance, the reactor mover comprising a first end, a first hinge, and a swing mechanism, wherein the swing mechanism comprises a swinging arm, a suspension frame constructed and arranged to at least partially embrace the reactor more than 180 degrees around the reactor, and a lift, wherein the first end is configured to mount to the frame of the apparatus, wherein the swing mechanism connects to the first end at the first hinge, wherein the lift is disposed on the swinging arm, and wherein the lift is constructed and arranged to move the reactor to allow for access to the reactor by a maintenance worker, and wherein the lift is constructed and arranged to move the suspension frame relative to the swinging arm.
2. The substrate processing apparatus according to claim 1, wherein the reactor mover is constructed and arranged to move the reactor horizontally to a maintenance area.
3. The substrate processing apparatus according to claim 2, wherein a displacer is constructed and arranged to move the reactor to the maintenance area outside a housing of the apparatus.
4. The substrate processing apparatus according to claim 3, wherein the apparatus is configured with the reactor mover to: lift the reactor up within the housing of the apparatus with the lift; move the reactor to the maintenance area outside the housing of the apparatus with the displacer; and, lower the reactor in the maintenance area outside the housing with the lift.
5. The substrate processing apparatus according to claim 2, wherein the first hinge is constructed and arranged to swing the reactor around a first substantially vertical axis.
6. The substrate processing apparatus according to claim 5, wherein the swing mechanism further comprises a second hinge constructed and arranged to swing the reactor around a second substantially vertical axis.
7. The substrate processing apparatus according to claim 6, wherein the swing mechanism further comprises a third hinge constructed and arranged to swing the reactor around a third substantially vertical axis.
8. The substrate processing apparatus according to claim 6, wherein the swinging arm is connected to the first hinge and the second hinge at opposite ends.
9. The substrate processing apparatus according to claim 1, wherein the lift is constructed and arranged to move the entirety of the suspension frame from a first vertical height to a second vertical height.
10. The substrate processing apparatus according to claim 1, wherein the swing mechanism is

constructed and arranged to swing around the first hinge.
11. The substrate processing apparatus according to claim 10, wherein the suspension frame comprises a C-shape.
12. The substrate processing apparatus according to claim 1, wherein the lift is provided with a linear guide mechanism comprising a linear guide, a carriage, a motor or a crank handle to support and move the reactor over the linear guide.
13. The substrate processing apparatus according to claim 12, wherein the linear guide is disposed along a second substantially vertical axis, wherein the first hinge is constructed and arranged to swing the swing mechanism around a first substantially vertical axis, and wherein the second substantially vertical axis is different than the first substantially vertical axis.
14. The substrate processing apparatus according to claim 12, wherein the linear guide mechanism forms part of the lift to move the reactor in a substantial vertical direction over the substantially vertical linear guide.
15. The substrate processing apparatus according to claim 1, wherein the reactor is mounted in the apparatus on a flange and the apparatus is constructed and arranged to clamp the flange to the reactor and the reactor mover is constructed and arranged to move the reactor together with the flange for maintenance.
16. The substrate processing apparatus according to claim 1, wherein the apparatus is provided with a first and second reactor and the reactor mover is constructed and arranged to move the reactor to a maintenance area provided in between the first and second reactor.
17. The substrate processing apparatus according to claim 1, wherein the swinging arm comprises a first swinging arm end and a second swinging arm end opposite the first swinging arm end, wherein the first hinge is connected to the first swinging arm end, and wherein the lift is constructed and arranged to move the suspension frame in a direction perpendicular to a direction between the first swinging arm end and the second swinging arm end.
18. The substrate processing apparatus according to claim 1, wherein the reactor mover comprises a suspension frame connected to a plurality of elongated suspensions, wherein the plurality of elongated suspensions are configured to suspend the reactor from the suspension frame with the plurality of elongated suspensions.
19. The substrate processing apparatus according to claim 18, wherein at least one of the plurality of elongated suspensions is aligned with a vertical direction.
20. A reactor mover for moving a reactor of a substrate processing apparatus for maintenance, wherein the reactor mover comprises a first end configured to mount to a frame of the substrate processing apparatus, a first hinge, and a swing mechanism comprising a swinging arm, a suspension frame, and a lift constructed and arranged to: move the reactor to a maintenance area outside the apparatus with a displacer; and, lower the reactor in the maintenance area outside the housing with the lift, wherein the first hinge connects the first end to the swing mechanism, wherein the swing mechanism is constructed and arranged to swing the swinging arm around the first hinge, wherein the lift is disposed on the swinging arm between the first hinge and the suspension frame, wherein the lift is constructed and arranged to move the suspension frame relative to the swinging arm, and wherein the suspension frame is configured and arranged to at least partially embrace the reactor more than 180 degrees around the reactor.
21. A method of maintaining a reactor of a substrate processing apparatus, the substrate processing apparatus comprising a frame on which the reactor is mounted, by: connecting a first end of a reactor mover to the frame of the substrate processing apparatus; connecting the reactor to a second end of the reactor mover, wherein the second end is on an opposite end of an arm of the reactor mover, and wherein the reactor is embraced more than 180 degrees around by the reactor mover; moving the reactor to a maintenance area outside the apparatus by swinging the reactor mover around a first hinge; lowering the reactor relative to the arm in the maintenance area outside a

housing with a lift, wherein the lift is disposed on the arm between the first hinge and the second end; and, providing maintenance on the reactor.